Eng., PhD., JUAN CARLOS OVIEDO CEPEDA

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Engineer with a Ph.D. in Electrical Engineering with an experience of nine years of research, five years of full-stack development, three years of building cloud-edge infrastructure and one year building agentic systems. At Hydro-Québec's research institute (IREQ) I architected and developed open-source cloud-edge platforms that turn buildings and distributed energy resources into autonomous grid assets, orchestrating real-time control across edge nodes. My work leverages machine learning on the edge to learn and predict thermal building state and manage the controllable loads to respond to grid needs. The control at the edge is carried out using machine learning, optimization or agentic systems to achieve the objectives of the distributed optimization problems formulated at the cloud level. I combine deep hands-on software engineering (Python, TypeScript, Kubernetes), IoT and communication capabilities (MQTT, Modbus, BACnet, OpenADR, etc.) with a scientist's grounding in control theory and convex optimization.

EXPERIENCE

Hydro-Québec | Research Scientist | 2021 – Present | 5 years

Systems Architecture

- Led the technical strategy and development of the OSED platform, a hybrid cloud-edge system for testing grid services like dynamic tariffs, distributed control (Python, Kubernetes, docker, MQTT, FastAPI, InfluxDB, TimescaleDB, ML, convex optimization, control, CI/CD, Grafana).
- Architected and developed a platform to measure the impact of autonomous AI agents on transactive energy markets (Python, docker, InfluxDB, Grafana).
- Prototyped the integration of the Bitcoin Lightning Network for real-time energy micropayments on the OSED platform, demonstrating my ability to rapidly build and test emerging technologies (Python, docker, InfluxDB, Grafana, LNBits).

Control & Optimization

- Created the Building Intelligence and the Predictive Control open-source libraries for optimizing energy assets (Python, ML, CVXPY, FastAPI, gRPC, Redis, TimescaleDB, MQTT, Zigbee, Modbus, Home Assistant).
- Developed and deployed applied machine learning models to learn building thermal dynamics and forecast energy consumption (Least Squares, CVXPY, SciPy, Pandas).
- Evaluated the impact of multiple Model Predictive Control strategies to define the best alternative for controlling buildings (Least Squares, CVXPY, SciPy, Pandas).

Smart Buildings & Demand Response

- Designed dynamic tariffs for reducing up to 21% the cost of energy on microgrids that power isolated communities.
- Created and presented the Open Grid-Client Laboratory in the Open Data Summit.
- Developed and tested a Home Energy Management System (HEMS) Proof of Concept, managing total household consumption (thermostats, water heaters, EV chargers, batteries) to respect power grid limits.

Agentic AI & Knowledge Graphs

- Architected and developed SI-MAPPER, a multi-modal Agentic AI solution that uses Computer Vision to parse engineering diagrams to generate structured ontological models (Ashrae 223P).
- Designed a foundational reference architecture that exposes real-time IoT data as a contextual service for Generative AI, enabling the rapid prototyping of LLM-based agents for smart grid applications.
- Re-architected a standard IoT control API into a Model Context Protocol server, enabling direct, natural language-based interaction with complex building energy systems (HVAC, EV chargers, batteries).

Data Science

- Conducted detailed analysis of electric vehicle impact on baseline consumption estimation, leading to valuable insights for demand response strategies.
- Analyzed errors in baseline consumption estimation by processing billions of data points across multiple substations, demonstrating expertise in Big Data analysis of real-world scenarios.

- Utilized Databricks, PySpark, and SQL to perform extensive analysis, showcasing proficiency in cutting-edge data processing technologies.

Research and Leadership

- Pioneered an open-source software strategy, leading [Hydro-Québec to join the Linux Foundation Energy], enhancing interoperability between data capture, AI agents, and IoT systems.
- Supervised students for research agreements between Hydro-Québec and the universities of Quebec at Trois-Rivieres and ETS, tutoring PhD students and postdocs on different subjects (+5 Ph.D. student interns, 1 Postdoc).

Contexto | Founder | 2025 – Present | 2 years

- Built and shipped a production MLOps entity resolution system on Azure ML mapping Colombian vehicle registry (RUNT) records to the SIBGA knowledge base via a 5-signal hybrid search pipeline (FAISS, BM25, fuzzy token-set, metadata, numeric/sinusoidal) over sentence-transformer embeddings, Optuna-tuned to 4.88% MAPE on price prediction; deployed through Azure DevOps CI/CD with weekly Scheduled-Pipeline + SendGrid health-check automation across online and batch endpoints. Stack: Python, Azure ML, Polars, FAISS, BM25, sentence-transformers, Optuna, Azure DevOps, SendGrid.
- Architected and shipped a multi-tenant AI video-generation SaaS where an async multi-step pipeline (script → scene blueprint → image → video) fans out per-scene jobs on Pub/Sub workers across Gemini, Veo, Kie AI, with CopilotKit-based agent orchestration for conversational script flows, BYOK per-account secrets in Secret Manager, Firestore real-time progress streaming, and Stripe-gated trial/access control at the API layer. Stack: TypeScript, Python, Next.js 16, FastAPI, Firebase Auth, Firestore, GCP (Cloud Run, Cloud Storage, Pub/Sub, Secret Manager), Stripe, CopilotKit, Gemini, Veo, Kie AI.
- Built a Next.js portfolio platform that models a professional career as a validated artifact graph, with end-to-end Sigma.js + Graphology rendering, and a multi-stage Docker build deployed to Cloud Run via Cloud Build. Stack: TypeScript, Next.js 16, React 19, Tailwind 4, Sigma.js, Graphology, ForceAtlas2, Firebase Auth, Firestore, Vitest, Playwright, Docker, GCP (Cloud Run, Cloud Build, Artifact Registry, IAM).

Université du Québec à Trois-Rivières | Adjunct Professor | 2023 – Present | 3 years

Research and Leadership

- Act as scientific co-director of 3 PhD students (Alejandro, Daniel, and Naren) at the Hydrogen Research Institute in the University of Quebec at Trois-Rivieres, the research focused on:
    1. Mechanism design for flexibility markets,
    2. Hierarchical clustering of users for increasing efficiency of grid services
    3. Economic efficiency comparison of Distributional Locational Marginal Prices and sharing the cost pricing schemes.
- Authored and co-authored 30+ academic papers with world-renowned experts.

EDUCATION

Formal education

- PhD in Electrical Engineering, Universidad Industrial de Santander, 2021. Received the distinction of honored thesis.
- B.S. in Electrical Engineering, Universidad Industrial de Santander, 2016. Graduated 4th out of 117 (top 3.4%).

Additional Courses & Certifications:

- From MLOps to LLMOps (Ivado), Design Thinking (MIT), Microeconomics (MIT), Blockchain and Money (MIT), Convex Optimization (Stanford), Game Theory (Yale).

AWARDS

- Winner of the IEEE BCTE competition for blockchain applications in energy (top 10 globally), 2022.
- Winner of the Emerging Leaders for the Americas Program (ELAP) scholarship to study in Canada.
- PhD thesis distinction: Honored thesis, 2021.
- Academic Excellence: Tuition fee exemption for three semesters, 2011-2012.